

Higher Order Spectra for Crack and Misalignment Identification in the Shaft of a Rotating Machine

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Higher order spectra (HOS) are the tools in signal processing for the identification of the presence of higher harmonics in a signal which is a typical case of a non-linear dynamic behavior in mechanical systems. The breathing of a crack during shaft rotation also exhibits a non-linear behavior. The crack is known to generate 2X (twice the machine RPM) and higher harmonics in addition to 1X component in the shaft response during its rotation. Misaligned shaft also shows such feature as a crack in a shaft. The HOS have now been applied on a small rotating rig to observe its features. Results are found to be encouraging to distinguish these two faults based on a few experiments conducted on a small rig, which are presented here. The presented results though limited suggest the potential of the use of the HOS in the condition monitoring of rotating machinery.

Keywords rotating machine · shaft crack · shaft misalignment · higher order spectra · bi-spectrum · tri-spectrum

1 Introduction

Faults keep generating in rotating machines. Identification of faults and their rectification is important from the safety and productivity points of view. Vibration based condition monitoring is generally used for such a requirement. Goldman and Muszynska [1] and Bosmans [2] gave the summary of the vibration based techniques for identifying different faults, and Sinha [3] gave a detailed spectrum of the vibration based condition monitoring. In general, each fault can be identified by some kind of characteristic responses of the shaft either during steady-state machine operation or during transient (run-up or

run-down) operation. For example, crack in a shaft or misaligned shaft generate 2X (twice the machine RPM) and higher harmonics in addition to 1X component in the shaft response during its rotation.

In the case of a crack in the shaft, the generation of the 2X component (twice the machine RPM) in the dynamic response of the shaft during its rotation is well known due to the variation in the shaft stiffness from high-to-low-to-high in a complete revolution of the shaft. This is because of the breathing (opening and closing) of the crack resulting from the rotor self-weight [4]. It is also known that both amplitude and phase of the 1X and 2X

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components change with the time of machine operation due to the propagation of the transverse crack. This can be observed in the polar plots (or the amplitude-phase versus time plots) [5] during the machine normal operation or the change in the orbit plot consisting of 1X and 2X components during the machine transient operation [6–8]. At the start of the machine the orbit plot changes from a single loop to double loops like a ‘figure eight’ with the increase in the speed, and then it changes to a loop containing a small loop inside when the speed is nearly half the natural frequency of the crack shaft indicating a significant change in the phase and amplitude of the 2X component [6–8]. At this particular moment, the shaft vibration may be significantly high to cause damage. Several researchers [9,10] have analytically shown such behavior of the orbit plots. The recent analytical simulation of the orbit plots of the responses of the crack shaft during machine transient operation by Sinou and Lees [11] highlights more precisely the behavior which is close to the experimental ones [3]. Wauer [12] and Gasch [13] gave the comprehensive review on the dynamics of cracked rotors. Sabnavis et al. [14] gave the latest advancement in the crack shaft detection and diagnosis.

The explanation for the existence of 2X component and higher harmonics in case of a misaligned shaft is not very straightforward (or not yet well understood) like the breathing of a crack. Ehrich [15] and Dewell and Mitchell [16] gave a review of the potential existence of 2X vibrations. In the simulation studies, a few researchers [17,18] have considered the forces and moments acting at the coupling due to misalignment in the shaft to explain the generation of 2X component. Sinha [3] and Sinha et al. [19] have used the constant forces and moments at the coupling together with the 1X responses to estimate the misalignment accurately in the experimental rig. Jordan [20] also confirms that the misalignment initially affects the 1X response resulting in an elliptical orbit but in the case of severe misalignment the orbit plot may look like a ‘figure eight’ due to the appearance of a high 2X component in the response. Hence it can be concluded that the exact phenomena for the generation of higher harmonics in case of

misalignment is not yet known. However it is known in practice that both the faults (crack and misalignment) can be identified and distinguished but often require transient operation of machine and the measurements in both horizontal and vertical directions in most cases.

The higher order spectra (HOS) – the bi-spectrum and the tri-spectrum are the tools for the detection of higher harmonics and their phase relationship in the signal [21–23]. They have also received the attention of the structural dynamic researchers to detect non-linear behavior in structural dynamic response, for example – detection of transient signal [24], looseness at the support [22,25], breathing of a crack [26], a cubical (non-linear) stiffness in the system [23,27], faults in the anti-friction bearings, cracks and gears, etc. [28,29], and the use of the HOS in the monitoring of the machining process in manufacturing [30], the early contact detection between two structural components, [34] etc. However none of the earlier studies used the HOS in the rotating machinery for the fault detection. Hence, the HOS have been computed for the shaft vibration responses for both the faults (crack shaft and misaligned shaft) simulated separately in the shaft of a small experimental rig. It has been observed that the HOS have the potential to identify and distinguish these two faults uniquely. The features of the HOS by the FE simulation of the cracked shaft experiment were also found to be similar to that of the experiments. The shaft response measurement in only one direction (either vertical or horizontal) maybe enough to identify these two faults, but requires more experimental observations. Hence the experimental results showing interesting observations when HOS were used have been presented here.

2 The HOS

The concept of the HOS is briefly discussed here.

2.1 Bi-spectrum

The conventional power spectrum density (PSD) provides information on the second-order

properties (i.e., energy) of a signal whereas the bi-spectrum can provide information on the signal's third-order properties. In a physical sense, the bi-spectrum provides insight into non-linear coupling between frequencies (as it involves both amplitudes and phases) of a signal compared to the traditional PSD that gives only the content of different frequencies and their amplitudes in a signal. The power spectrum of a time series $\mathbf{x}(t)$ is computed by the discrete Fourier transform (DFT) of the signal as

$$\text{PSD, } \mathbf{S}_{xx}(f_k) = E[\mathbf{X}(f_k)\mathbf{X}^*(f_k)], \quad k = 1, 2, 3, \dots, N \quad (1)$$

where $\mathbf{S}_{xx}(f_k)$ is the PSD, $\mathbf{X}(f_k)$ and $\mathbf{X}^*(f_k)$ are the DFT and its complex conjugate at frequency f_k for the time series $\mathbf{x}(t)$. N is the number of the frequency points. The bi-spectrum is the double Fourier transformation of the third-order moment of a time signal [21,23] that involves two frequencies components (both amplitudes and phases) of the signal, and computed by the signal DFT as [23]

$$\text{Bi-spectrum, } \mathbf{B}_{xxx}(f_l, f_m) = E[\mathbf{X}(f_l)\mathbf{X}(f_m)\mathbf{X}^*(f_l + f_m)], \quad l + m \leq N. \quad (2)$$

The bi-spectrum is complex and interpreted as measuring the amount of coupling between the frequencies at f_l , f_m , and $f_l + f_m$, and is described by 'quadratic phase coupling'.

2.2 Tri-spectrum

Similarly, the tri-spectrum is the triple Fourier transformation of the fourth-order

moment of a time signal, and computed as [23]

$$\begin{aligned} \text{Tri-spectrum, } \mathbf{T}_{xxxx}(f_l, f_m, f_n) \\ = E[\mathbf{X}^*(f_l)\mathbf{X}^*(f_m)\mathbf{X}^*(f_n)\mathbf{X}(f_l + f_m + f_n)], \\ l + m + n \leq N. \end{aligned} \quad (3)$$

The concept of correlating different harmonic components (both amplitudes and phases) in the HOS clearly indicates that they are the non-linear estimators. It is because the non-linear behavior in the structural dynamics generates several harmonics of the exciting frequencies.

3 Experiments

The schematic of the test rig is shown in Figure 1. The rig consists of a 10mm OD steel shaft of 550mm long supported on two bush bearings, which are directly mounted on a rigid massive base plate. The left end of the shaft is coupled with the motor shaft through a flexible coupler. The bearings are placed at 20mm and 510mm from the left end of the shaft. The shaft also carries one balance disk made of steel and placed at the midspan of the two bearings. Balance disk dimensions are 75mm OD, 10mm ID, and 25mm thickness. The first natural frequency of the shaft (with no crack) in the rig is identified at 27.50 Hz (both horizontal and vertical directions) by the modal tests using the impulse-response method [31].

Most of the rotating machines, e.g., turbo-generator (TG) sets are now equipped with the proximity probes near the bearings to monitor the shaft relative displacements during normal operation. Here also proximity probes both

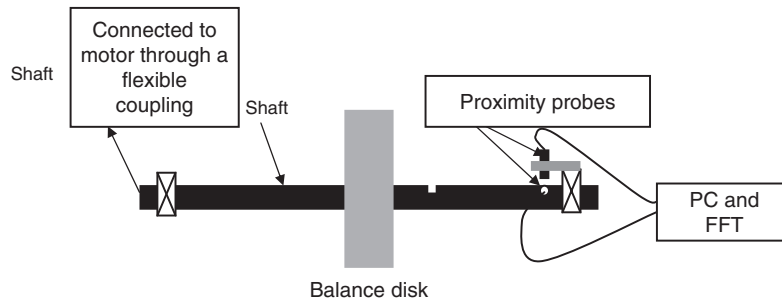


Figure 1 Schematic of the experimental rig.

in vertical and horizontal directions were mounted on the bearing-2 housing of the rig to measure the shaft displacement so that the observations made here may be useful for the real life machines. These arrangements are also shown in Figure 1.

Experiments [3] were carried out on the rig for both faults separately – one with a crack in the shaft and other with misaligned shaft. The data were recorded in the PC with the sampling rate of 2560 samples/s with anti-aliasing filter of 1 kHz for both experiments. The experimental observations in both cases are discussed below. It may be important to note that the only prominent 1X component in the measured responses was observed for the rig without these faults.

3.1 Fault 1: Cracked Shaft

The crack in the shaft was developed by perfectly welding the two pieces of a shaft such that the breathing of the crack can be possible during the shaft rotation. The location of the crack was at 315 mm from left and depth equals to half of the shaft diameter. The first natural frequency of the cracked shaft is identified at 26.25 Hz in the vertical direction when the crack was fully open by modal tests [31].

Experiments were conducted at different rotating speeds during machine run-up to explore the behavior of crack breathing during the shaft rotation. A change in the orbit behavior during the machine run-up from a single loop at low speed to a double loop to a loop containing a small loop at 750 RPM (nearly half the natural frequency of crack shaft) indicates a significant change in the phase and amplitude of the 2X component was observed as expected for a crack shaft [3]. These observations are well known and generally used in the vibration based condition monitoring [6]. The two typical speeds – 650 and 750 RPM were only considered here as at these two speeds significant changes were seen in inter-phase relations between harmonics. This is evident from the vibration spectra and the orbit plots at these two shaft RPM in Figures 2 and 3. The vibration was very severe at 750 RPM, and hence the speed was not increased further during the experiment.

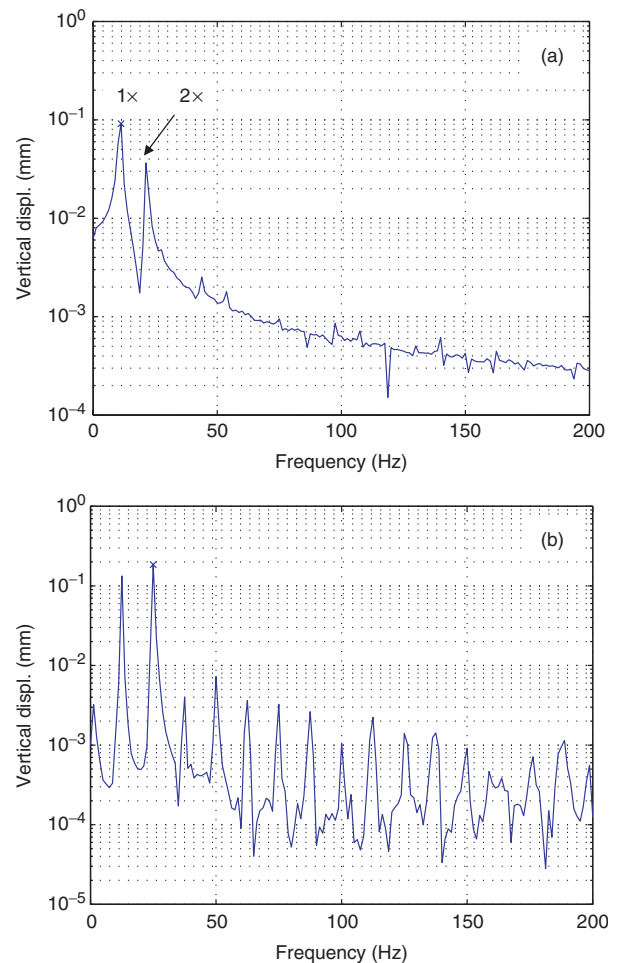


Figure 2 The amplitude spectra of the cracked shaft in vertical direction: (a) 650 RPM and (b) 750 RPM.

3.2 Fault 2: Misaligned Shaft

In the same rig, a misalignment was introduced in the shaft without crack by disturbing the alignment between two bearing pedestals. The bearing pedestal near the flexible coupling to the motor shaft has been misaligned by 1 mm and 0.5 mm in the vertical and horizontal directions, respectively. This arrangement has introduced misalignment effect due to the coupling of the rotor with the motor shaft. The measurements were then carried out at the shaft speed starting from 300 to 900 RPM at the steps of 150 RPM to bring out the difference of behavior in the HOS. A few typical amplitude spectra of the misaligned shaft responses in the vertical direction are shown in Figure 4.

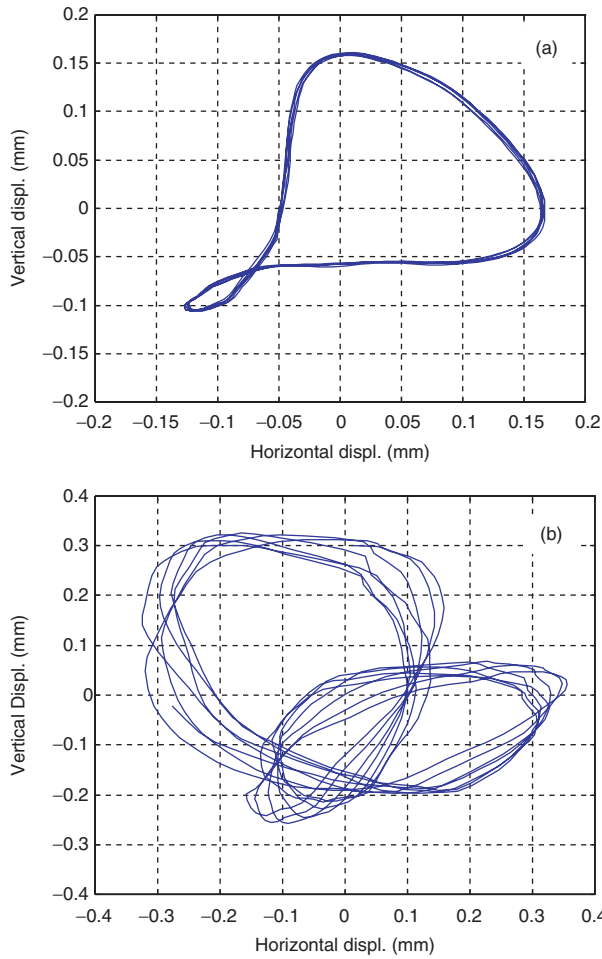


Figure 3 The orbit plots of the cracked shaft responses: (a) 650 RPM and (b) 750 RPM.

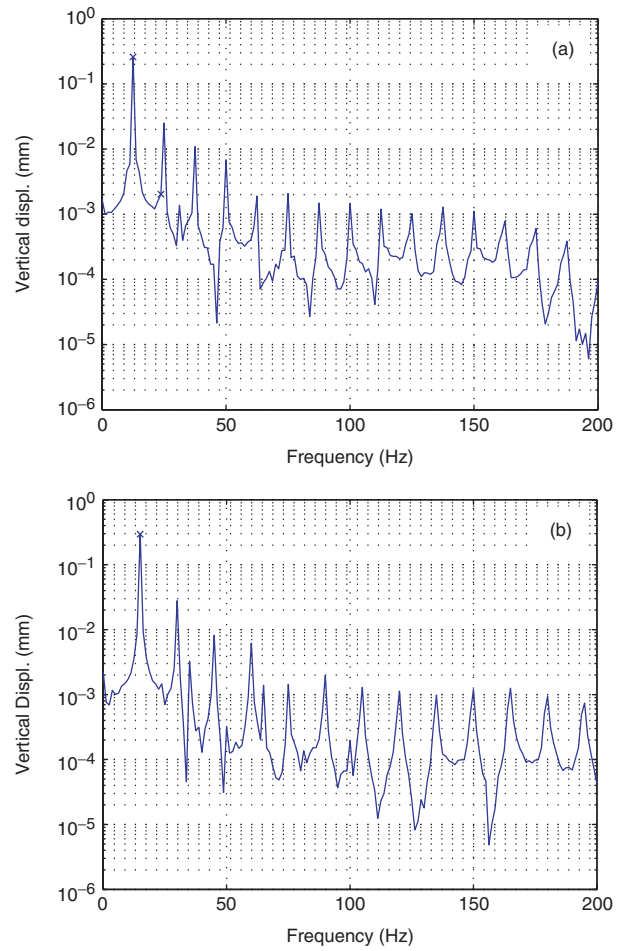


Figure 4 The amplitude spectra of the misaligned shaft in vertical direction: (a) 750 RPM and (b) 900 RPM.

3.3 The HOS for Both the Faults

Here both bi-spectrum and tri-spectrum were computed for the measured responses at the different speeds for both the faults. The 50 segments using 50% overlap in the time data with the frequency resolution of 1.25 Hz were used for the averaging of the computed HOS (bi-spectra and tri-spectra) using Equations (2) and (3). If the machine run-up duration is very short then the general interpolation FFT method suggested by Shi et al. [32] can be utilized to construct a long duration response at a desired machine speed. A few typical spectra (HOS) are shown in Figures 5–8. The amplitude is normalized to 1 for both the bi and tri-spectra. The diameter of the ball indicates the amplitude

of a tri-spectrum related to the three frequency axes – f_i , f_m , and f_n . Only the amplitudes above 0.1 in the tri-spectrum are plotted here.

4 Discussion

Although the amplitude spectra contain several higher harmonics in the response for both the crack and misaligned shaft responses, a total of four peaks are only seen in the bi-spectra, shown in Figure 5, for the crack shaft. Since the crack in the structures produced multiple harmonics of the exciting frequency, the viewing of the bispectrum results in terms of their components like $\mathbf{B}_{11}$, $\mathbf{B}_{12}$, $\mathbf{B}_{22}$, etc. would be useful.

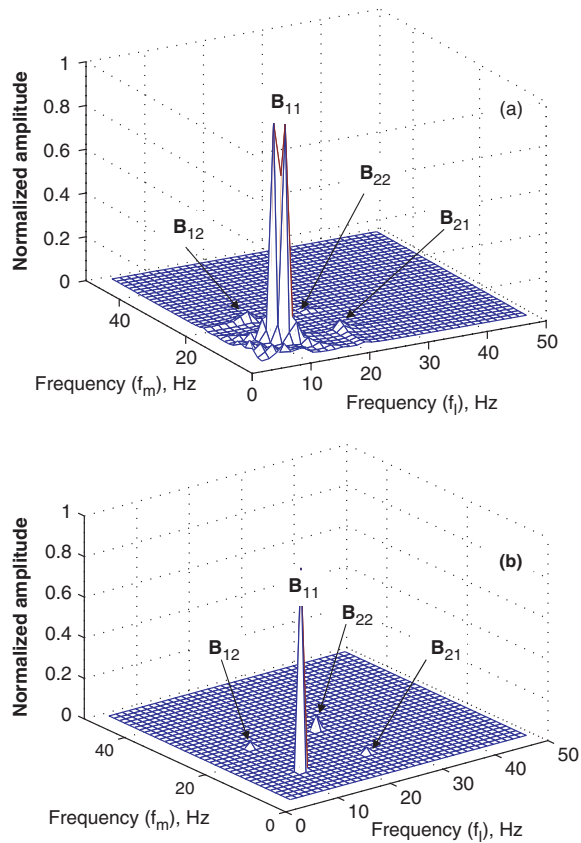


Figure 5 The bi-spectra of the cracked shaft: (a) 650 RPM and (b) 750 RPM.

The diagonal peaks in a bi-spectrum are indicative of the relation between two consecutive harmonics of a spectrum. Hence, the peak $\mathbf{B}_{11}$ relates 1X and 2X components and $\mathbf{B}_{22}$ relates 2X and 4X components of a spectrum. The off-diagonal peak $\mathbf{B}_{12}$ ($=\mathbf{B}_{21}$) relates responses at the three harmonic frequencies – 1X, 2X, and 3X. The peak $\mathbf{B}_{22}$ started appearing at 650 RPM when the orbit behavior changes from a double loop to a loop containing a small loop at the 750 RPM [3]. Whereas a total of five peaks has always appeared in the bi-spectra shown in Figure 6 irrespective of the rotor speed in the case of the misaligned shaft. An additional off-diagonal peak $\mathbf{B}_{13}$ ($=\mathbf{B}_{31}$) observed in the misaligned shaft response gives the relation between the 1X, 3X, and 4X components of the spectrum. The non-existence of $\mathbf{B}_{22}$ and the similar feature

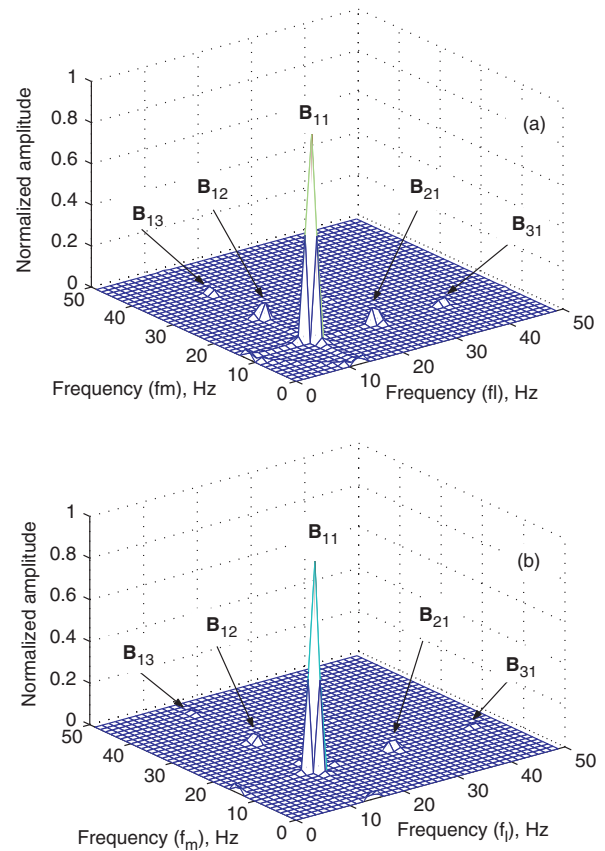


Figure 6 The bi-spectra of the misaligned shaft: (a) 750 RPM and (b) 900 RPM.

at all speeds indicate that the misalignment is not a speed dependent phenomenon and the phase relation between the 2X and 4X components is certainly much different from the cracked shaft.

Similarly, the tri-spectra for the crack shaft response are speed dependent and contain several tri-spectral components ($\mathbf{T}_{111}, \mathbf{T}_{112}=\mathbf{T}_{121}=\mathbf{T}_{211}, \mathbf{T}_{122}=\mathbf{T}_{212}=\mathbf{T}_{221}$, and $\mathbf{T}_{222}$) with their amplitudes depending on the operating speed as shown in Figure 7. However the only component of $\mathbf{T}_{111}$ has been observed in the tri-spectra plots in Figure 8 for the misaligned shaft response irrespective of the machine speed.

The observations on the bi-spectra and tri-spectra behavior clearly indicate that the shaft responses may have several higher harmonics related to the machine speed in their amplitude spectra but the phase relations among them are

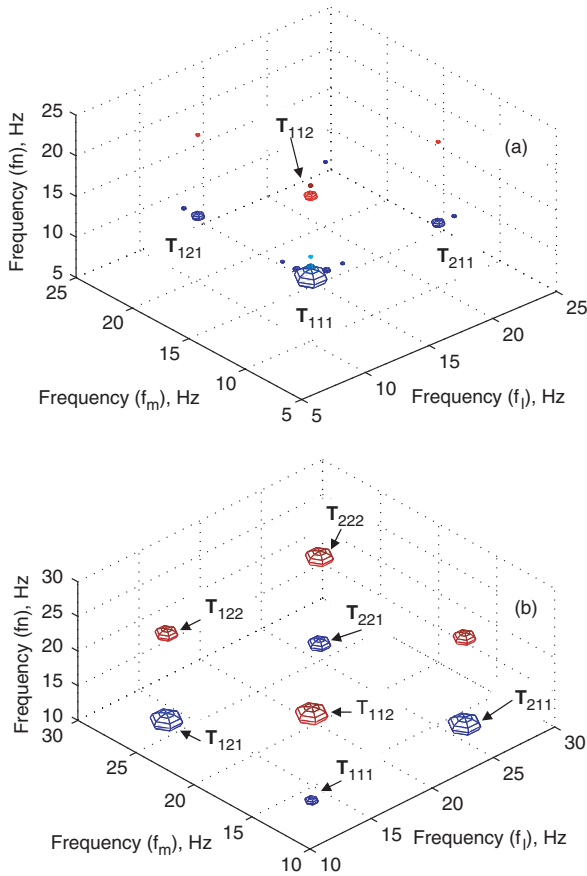


Figure 7 The tri-spectra of the cracked shaft: (a) 650 RPM and (b) 750 RPM.

significantly different, which can be brought out through the HOS analysis. The phase information may be useful for distinguishing the different faults as two of them are demonstrated here. In fact, the amplitude of the bi-spectrum and tri-spectrum components for the responses at different speeds of the shaft without any faults were found to be <0.3 indicating the non-existence of higher harmonics, and the noise content at all frequency components than the shaft RPM. Hence the appearance of different components in the bi-spectrum and tri-spectrum for both faults – crack and misalignment; however, several experiments need to be conducted on relatively bigger rigs with different small crack sizes, and different levels of misalignment to bring out the robustness of the present observation.

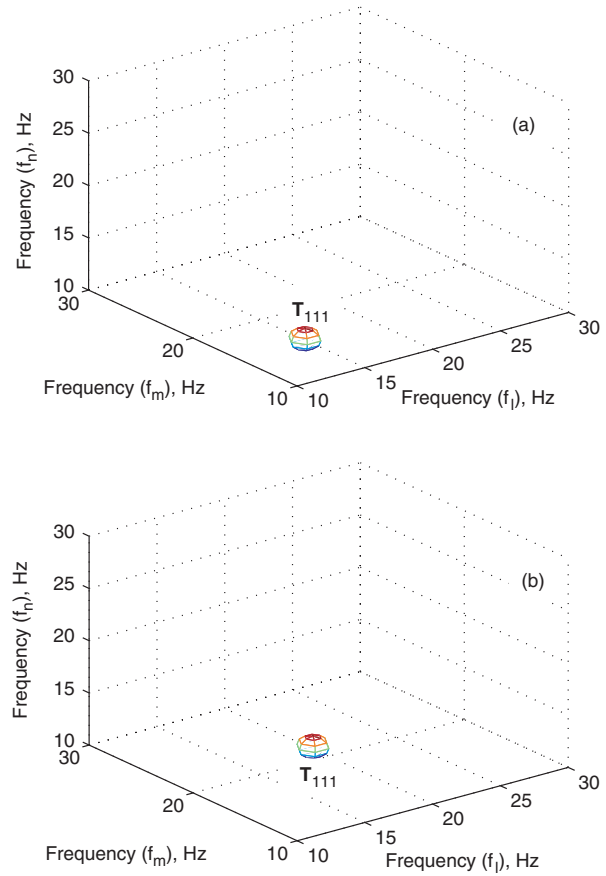


Figure 8 The tri-spectra of the misaligned shaft: (a) 750 RPM and (b) 900 RPM.

5 Finite Element Simulation

A finite element simulation of the experiment was also carried out. A finite element model was created for the rotor using two-noded Euler–Bernoulli beam elements, each with two translational and two rotational degrees of freedom. The first two natural frequencies calculated are 26.53 and 228.62 Hz in both the vertical and horizontal directions. Then the crack was introduced in the FE model using the method suggested earlier [33]. The calculated natural frequencies in the vertical (fully open crack) and horizontal directions are 25.75 and 225.68 Hz, and 26.10 and 226.98 Hz, respectively. The calculated frequencies are close to the experimentally measured values. The stiffness proportional damping matrix was also included

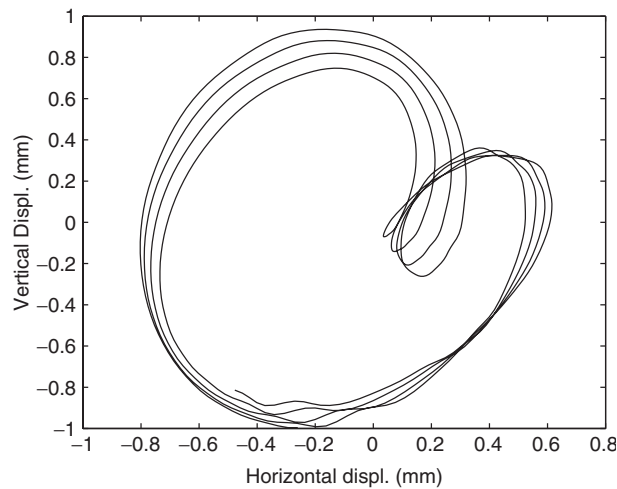


Figure 9 The orbit plot of the cracked shaft estimated by FE analysis.

in the model using experimentally measured modal damping of 0.3% at the first mode. To this model, the unbalance responses in vertical and horizontal directions were estimated assuming small unbalance at the disk. A simple model for the breathing of crack (completely open during vertically lower position and completely closed for vertically top position for the crack in a rotation) was assumed for the response estimation. The crack breathing model was a function like $(1 - \cos\theta)\Delta k/2$ in the equation of motion, where θ is the angle of shaft crack rotation from the closed position, and Δk is the maximum reduction in the shaft stiffness when crack is completely open. The responses were then estimated when the shaft was assumed to be rotating at the speed of half the first natural frequency i.e., 772.5 RPM. The Newmark- β method was used for the numerical solution of the system dynamic equations. A time step 1/10000s was used for the numerical integration. The estimated displacement responses were low pass filtered at 1 kHz, and then data were down sampled by 10 so that the sampling frequency becomes 1 kHz the same as experiments. Their responses were then polluted by the random noise such that the SNR becomes 40dB before the computation of the HOS. Figure 9 gives the orbit plot and Figure 10 the bi-spectrum and the tri-spectrum for the estimated responses at the measured locations which are close to the experimental

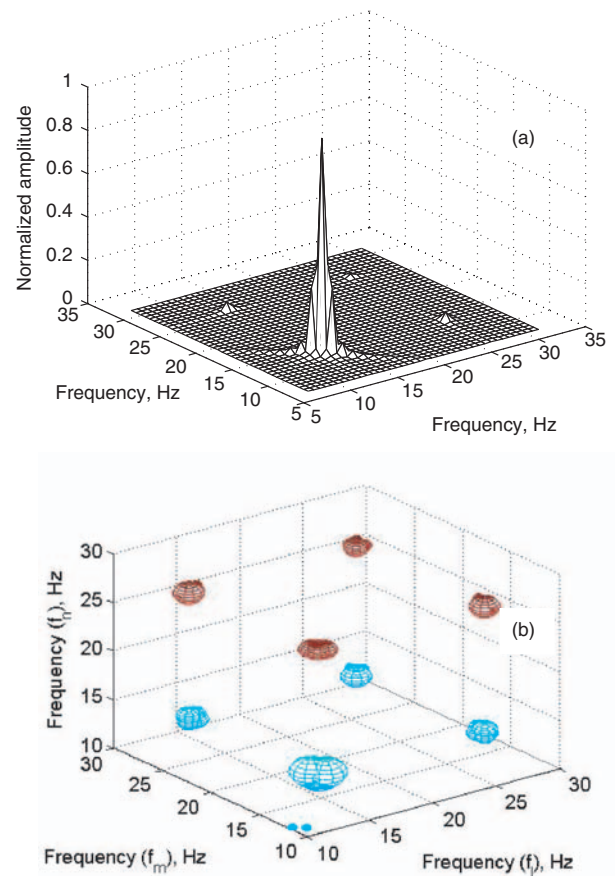


Figure 10 The HOS of the cracked shaft estimated by FE analysis: (a) bi-spectrum and (b) tri-spectrum.

observations. Similar simulation for the misaligned shaft could not be done, as the force function due to the shaft misalignment is not well stood.

6 Conclusions

In machine condition monitoring the detection of faults in time is utmost important for the critical rotating machines in a plant. The bi-spectrum and tri-spectrum have been applied first time in the condition monitoring of the rotating machinery, and they have shown encouraging results to identify and distinguish two faults (crack and misalignment) based on the limited initial experiments on a small rig. Based on the initial observations, it seems that the shaft response measurement in a direction (either vertical or horizontal) may be capable to identify

these faults. It may be useful in reducing the number of instrumentation needed at present for the condition monitoring. But it requires more experiments on relatively big rigs to enhance the confidence level in the diagnosis. At present only two faults – crack with relatively big size and misalignment were simulated in the rig, these need to be verified for a small crack size in relatively big diameter shaft as well as both faults exists simultaneously. Furthermore, similar simulations need to be done for other types of faults so that the usefulness of the HOS in a machine condition monitoring can be completely explored. This is being planned as future study.

Acknowledgments

The small test rig facility at BARC, Mumbai India used for the experiments by the author in year 2002 during his PhD study is gratefully acknowledged.

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